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MILITARY SPECIFICATION

KETTLES, STEAM-JACKETED (STAINLESS STEEL), ELECTRICALLY HEATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrically heated, stainless steel, steam-jacketed kettles.

1.2 Classification. Steam-jacketed kettles covered by this specification shall be of the following types and sizes as specified (see 6.2 and 6.6).

Type I - Floor model:
  Sizes:
    Size 1 - 20-gallon (75.7 Liters [L]) capacity
    Size 2 - 30-gallon (113.6 L) capacity
    Size 3 - 40-gallon (151.4 L) capacity
    Size 4 - 60-gallon (227.1 L) capacity
    Size 5 - 80-gallon (302.8 L) capacity

Type II - Table top model (permanent water supply):
  Sizes:
    Size 6 - 2-1/2-gallon (9.5 L) capacity
    Size 7 - 5-gallon (18.9 L) capacity
    Size 8 - Two (twin) 5-gallon (18.9 L) capacity
    Size 9 - 10-gallon (37.9 L) capacity

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7310

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1.2.1 Part numbers. The specification number, type, and size are combined to form the specification part numbers for the kettles covered by this specification. The part numbers for the kettles are established as follows:

M43359 XXX

Specification number ____
Type and size code (see 1.2.1.1) ___

1.2.1.1 Type and size code. The type and size code is identified by a 3-digit number. The first digit identifies the type and the last two digits identifies the size as shown in table I.

TABLE I. Type and size to code number.

Types and Sizes	Code Number
TYPE I	
Size 1	101
Size 2	102
Size 3	103
Size 4	104
Size 5	105
TYPE II	
Size 6	206
Size 7	207
Size 8	208
Size 9	209

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

RR-T-25 - Table, Steam-Jacketed Kettle.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-60 - Tape: Packaging, Waterproof.
PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

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MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Section VIII - Unfired Pressure Vessel Code.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
A240 - Stainless and Heat-Resisting Chromium and Chromium-Nickel Steel Plate, Sheet, and Strip for Fusion-Welded Unfired.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS 1 - General Standard for Industrial Controls and Systems.
ICS 2 - Industrial Control Devices, Controllers and Assemblies, Standards for.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York, NY 10017.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 - National Electrical Code.

(Applications for copies should be addressed to the National Fire Protection Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL SANITATION FOUNDATION (NSF)

No. 2 - Food Service Equipment.
No. 4 - Commercial Cooking and Warming Equipment.
Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 197 - Commercial Electric Cooking Appliances.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard commercial product. The kettle shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the kettle being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection and approval (see 4.2.1 and 6.4).

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3.3 Codes and standards. The kettles shall conform with the requirements of UL 197, ASME Unfired Pressure Vessel Code - Section VIII (hereinafter referred to as the ASME Code) NSF No. 4, NEMA ICS 1, and NEMA ICS 2. The stands and tables shall comply with NSF No. 2.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4.1 Stainless steel. Stainless steel shall conform to the 300 series of ASTM A167 or ASTM A240, as applicable.

3.4.2 Insulation. Insulation shall be commercial grade fiberglass blanket or blocks not less than 1-inch (25.4 millimeters [mm]) thick.

3.4.3 Pipe. Pipe shall be Iron Pipe Size (IPS) with National Pipe Taper (NPT) threaded ends. Pipe for water lines shall be seamless brass, standard weight.

3.4.4 Pipe fittings. Pipe fittings used with brass pipe shall be bronze, and of the type and weight normally furnished by the manufacturer on his standard product. Pipe fittings welded to the steam-jacket shall be stainless steel.

3.5 Design.

3.5.1 Type I. Type I kettles shall be stainless steel and shall be designed to hold not less than 20 gallons (75.7 L), 30 gallons (113.6 L), 40 gallons (151.4 L), 60 (227.1 L), or 80 gallons (302.8 L) of liquid as specified (see 6.2). The kettles shall be designed to generate steam by means of an electric element. Kettles shall have a working pressure of not less than 10 pound-force per square inch (psi) (68.94 kilo Pascals [kPa]) and shall operate at any steam-jacket pressure from vacuum up to their rated working pressure. Kettles shall consist of an inner body with a steam-jacket, and when required for Navy shipboard use (see 3.6.1.3 and 6.2), shall be enclosed by insulation and an outer casing. Kettles shall have provisions for filling the steam-jacket at installation.

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3.5.2 Type II. Type II kettles shall be designed to hold not less than 2-1/2 gallons (9.5 L), 5 gallons (18.9 L), two (twin) 5 gallons (18.9 L for each kettle), and 10 gallons (37.9 L) of liquid as specified (see 6.2). Kettles shall be designed for a permanent supply of water and shall generate steam by means of an electric element. Unless otherwise specified (see 6.2), the type II kettles shall be charged at the factory with a permanent supply of water containing rust inhibitors and any other necessary additives. The working pressure shall be not less than 10 psi (68.94 kPa) and the kettle shall operate at any steam-jacket pressure from vacuum up to rated working pressure. Kettles shall primarily consist of an inner body with a pouring lip, and controls and safety devices necessary for safe, efficient operation. The kettle shall be designed for table top, counter top, or stand mounting.

3.5.3 Electrical characteristics. Unless otherwise specified (see 6.2), the heating elements, contactors, and other electrical components of the kettles shall be designed to operate on the electrical parameters as specified in table II. When the kettles are specified for Navy shipboard use (see 6.2), the heating elements, contactors, and other electrical components of the kettles shall be designed to operate on 440 volts (V) alternating current balanced 3-phase. Provisions shall be made for grounding the kettles. Wiring practices shall conform to NFPA 70.

3.5.4 Repair and maintenance. Provisions shall be made to allow for ready service and replacement of controls. In addition, there shall be free access to inspect, service, and adjust any component or equipment as provided for by this specification or as indicated by the component or equipment manufacturer, without requiring disassembly of any other major component or equipment on the kettle.

TABLE II. Electrical requirements.

Types and sizes	Operating voltages 1/ (nominal)	Wattage ratings of heating elements (minimum)
Type I		
Size 1, 20 gallon (75.7 L)	208 V, 60 Hertz (Hz) 3-phase	9 kiloWatts (kW)
Size 2, 30 gallon (113.6 L)	208 V, 60 Hz, 3-phase	12 kW
Size 3, 40 gallon (151.4 L)	208 V, 60 Hz, 3-phase	14.8 kW
Size 4, 60 gallon (227.1 L)	208 V, 60 Hz, 3-phase	21 kW
Size 5, 80 gallon (302.8 L)	208 V, 60 Hz, 3-phase	24 kW

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TABLE II. Electrical requirements. (continue)

Types and sizes	Operating voltages 1/ (nominal)	Wattage ratings of heating elements (minimum)
Type II Size 6, 2.5 gallons (9.5 L)	208 V, 60 Hz, single phase	3 kW
Size 7, 5 gallons (18.9 L)	208 V, 60 Hz, 3-phase	6 kW
Size 8, two (twin) 5 gallon kettles (18.9 L)	208 V, 60 Hz, 3-phase	12 kW (6 kW for each 5 gallon kettle)
Size 9, 10 gallon (37.9 L)	208 V, 60 Hz, 3-phase	9.8 kW

1/ Unless 440 V, balanced 3-phase is specified for Navy shipboard use (see 6.2).

3.6 Construction. The kettles shall be constructed as specified herein. Unless otherwise specified herein stainless steel shall be type 304 or type 304L where parts are fabricated by welding.

3.6.1 Type I kettles. Type I kettles shall conform to the dimensional requirements of table III.

TABLE III. Dimensions (type I).

Measurements	20 gallons (75.7 L)	30 gallons (113.6 L)	40 gallons (151.4 L)	60 gallons (227.1 L)	80 gallons (302.8 L)
	Inches (mm)	Inches (mm)	Inches (mm)	Inches (mm)	Inches (mm)
Maximum overall outside diameter at kettle top	28-1/2 (723.9)	30-1/2 (774.7)	32-1/2 (825.5)	36 (914.4)	39 (990.6)
Maximum height from floor to kettle top	37-3/4 (958.8)	38-1/2 (997.9)	40 (1016.0)	43 (1092.2)	46 (1163.4)

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TABLE III. Dimensions (type I). (continue)

Measurements	20 gallons (75.7 L)	30 gallons (113.6 L)	40 gallons (151.4 L)	60 gallons (227.1 L)	80 gallons (302.8 L)
Minimum clearance between floor and draw-off outlet	12 (304.8)	12 (304.8)	12 (304.8)	12 (304.8)	12 (304.8)
Minimum swing spout water valve and spout size (IPS)	1/2 (12.7)	1/2 (12.7)	1/2 (12.7)	1/2 (12.7)	1/2 (12.7)
Water compartment inlet connection and globe valve (IPS) (when provided)	1/2 (12.7)	1/2 (12.7)	1/2 (12.7)	1/2 (12.7)	1/2 (12.7)
Minimum drain for steam-jacket (IPS)	3/8 (9.65)	3/8 (9.65)	3/8 (9.65)	3/8 (9.65)	3/8 (9.65)

3.6.1.1 Kettle body. The body shall be cylindrical with either a hemispherical or sloping bottom and shall be fabricated from type 302 or 304 stainless steel as specified in 3.4.1. The material thickness of that section of the body forming the inner wall of the steam-jacket shall conform to the requirements of the ASME Code for the rated working pressure of the kettle. The body shall be formed seamless from one piece of material or die formed and welded in sections. The top edge shall be flanged or welded to form a canopy and shall slope away from the body. The rim of the body shall fit the flange on the cover. The bottom of the body shall be provided with an opening to the draw-off assembly and shall be located to allow complete drainage.

3.6.1.2 Steam-jacket. The outer wall and bottom of the steam-jacket shall be stainless steel. Material thickness shall conform to the requirements of the ASME Code for the rated working pressure of the kettle. The steam-jacket shall be formed of one piece or welded in segments, and the top shall be welded to the body. The draw-off tube shall pass through the steam-jacket and shall be fully welded where it exits to form a leakproof joint. The steam-jacket shall be provided with a low water cutoff, safety valve, pressure gauge, and a means for filling and adding of water to the steam-jacket. Provisions shall be made for complete drainage of the steam-jacket.

3.6.1.3 Casing. When specified (see 6.2), the body and steam-jacket shall be enclosed by a cylindrically shaped casing designed so that it can be removed without breaking any welds. The top of the casing shall terminate at the canopy and shall be attached thereto, and the bottom shall terminate below the steam-jacket. The casing shall consist of one outer liner or separate

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inner and outer liners. Insulation as specified in 3.4.2 shall be applied between the kettle body and outer lining or between the two linings, as applicable. The inner liner shall be formed from not less than 0.045 inch (1.14 mm) thick stainless steel or not less than 0.0538 inch (1.37 mm) thick carbon steel sheet or electro-galvanized metal, and the outer liner shall be formed from not less than 0.045 inch (1.14 mm) thick stainless steel sheet as specified in 3.4.1. When an inner and outer lining is provided, carbon steel reinforcing rings located at the top and bottom shall separate the inner and outer liners. The reinforcing rings shall be fabricated from not less than 1-inch by 1-inch by 1/8-inch (25.4 by 25.4 by 3.2 mm) angle, rolled to the radius of the outer casing and welded to the inner and outer casings. The stainless steel canopy specified in 3.6.1.1 shall center the kettle within the casing. The top of the outer lining shall be attached to the canopy by stainless steel sheet metal screws.

3.6.1.4 Legs. The kettles shall be mounted on not less than three stainless steel legs with adjustable feet. Foot adjustment shall be not less than 1/2 inch (12.7 mm).

3.6.1.5 Covers. Covers shall be formed from type 302 or 304 stainless steel as specified in 3.4.1 and shall be a nominal 0.050 inch (1.27 mm) thick or thicker. The covers shall be flanged, dished, or suitably reinforced to prevent distortion or permanent set as a result of normal use. The covers shall fit snugly to the top of the kettle body. All covers for the type I kettle shall be provided with means to insure that condensation returns to the kettle interior.

3.6.1.5.1 Covers for type I kettles up to 40 gallons (151.4 L). The 20 gallon (75.7 L), 30 gallon (113.6 L), and 40 gallon (151.4 L) kettles shall be provided with a one-piece hinged cover. The hinges shall be the offset-type located at the back of the kettle. The cover shall be designed to prevent distortion as a result of normal use, and shall be provided with a handle as specified in 3.6.1.5.3. The cover shall open at not less than 65 degrees (°) (1.1 radians [rad]) from the closed position and shall have provisions to retain the cover in the open position.

3.6.1.5.2 Covers for type I kettles over 40 gallons (151.4 L). The 60 gallon (227.1 L) and 80 gallon (302.8 L) kettles shall be provided with either a one-piece hinged cover as specified in 3.6.1.5.1 or a standard manufacturers two-piece hinged cover. The two-piece cover shall be provided with two offset hinges at the back of the kettle and two hinges located between the back of the kettle and the centerline of the kettle. Hinges shall be secured to the body and cover with either a continuous filled weld or spot welds. The cover shall be designed to prevent distortion or permanent set as a result of normal use. The two sections of the cover shall open not less than 100° (1.75 rad) from the closed position and shall rest in this position through the medium of stops provided on the back of the hinges. A sanitary shield shall be located at the rear of the front cover section, and the front edge of the back section shall be flanged-up to prevent dirt and foreign matter from entering the kettle through the hinged joints.

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3.6.1.5.3 Cover handle. Each cover shall be provided with a raised handle made of stainless steel bar stock or tubing. A plastic knob may be provided on the handle. The handle shall be located so the cover can be raised without the attendant's arm passing over the open kettle. The handle shall be secured to the cover by spot-welding or fillet-welding.

3.6.1.6 Draw-off assembly. The kettles shall be provided with a draw-off assembly located tangent with the lowest point inside the body. The assembly shall consist of the draw-off tube as specified in 3.6.1.6.1 and the draw-off valve as specified in 3.6.1.6.2. The draw-off tube shall have an outside diameter of not less than 1-1/2 inches (38.1 mm) and the draw-off valve shall be nominal 1-1/2 inch (38.1 mm) size. When specified (see 6.2), the draw-off tube and draw-off valve for the type I kettles shall be nominal 3-inch (76.2 mm) diameter and size, respectively.

3.6.1.6.1 Draw-off tube. The draw-off tube shall be formed of stainless steel tube and shall be welded tangent to the lowest point of the body and to the steam-jacket where the draw-off tube pierces it. The draw-off tube shall maintain the same section throughout its entire length and shall be smooth on the inside to prevent the lodgement of food particles. The inside of the draw-off tube shall be flush with the round opening in the draw-off valve. The outlet of the draw-off tube shall be provided with a removable strainer fabricated of stainless steel as specified in 3.4.1. The stainless steel shall be perforated with nominal 1/4 inch (6.354 mm) diameter holes located on not greater than 9/16 inch (14.29 mm) centers and shall be welded to a stay which shall extend downward and fit snugly into the outlet tube. A stainless steel strainer hook with a loop or tee handle shall be provided for removal of the strainer and shall be made from nominal 3/16 inch (4.8 mm) diameter rod, 6 inches (152.4 mm) longer than the inside depth of the kettle.

3.6.1.6.2 Draw-off valve. The draw-off valve shall be a sanitary, angle, compression-disc type fabricated of nickel alloy dairy metal or number 304 stainless steel. The draw-off valve shall be soldered to the draw-off tube by means of a recessed sanitary joint or may be bolted to a face plate. The stem for the 1-1/2 inch (38.1 mm) draw-off valve shall be provided with a fiber-impregnated plastic handle not less than 2-1/8 inches (54 mm) in diameter or a bar type handle not less than 5 inches (127 mm) in length. The 3 inch (76.2 mm) valve shall have a wheel type metal handle of not less than 5 inches (127 mm) in diameter or a bar type handle of not less than 6 inches (152.4 mm) in length. The draw-off valve shall be so designed that it can be taken apart without the use of tools. "V" or modified "V" threads shall not be used in the draw-off valve.

3.6.1.7 Swing-spout. Unless otherwise specified (see 6.2), a swing-spout water supply connection shall be provided. The spout shall be a combination swing-spout and faucet or a swing-spout with globe valve as specified in 3.6.1.7.1. The spout shall be the manufacturer's standard design and shall be chrome-nickel-plated brass or stainless steel. The assembly shall be mounted on a not less than 0.0565 inch (1.4 mm) thick stainless steel bracket which shall be securely attached to the kettle. The bracket shall be positioned on

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the right side of the kettle when facing the kettle draw-off valve. The bracket and swing-spout shall be so positioned that the outlet end of the spout has not less than a 1-1/2 inch (38.1 mm) clearance above the top rim of the kettle. When a faucet is not an integral part of the swing-spout, a globe valve of the same size as the water inlet connection shall be inserted directly below the swing-spout bracket assembly. The spout outlet shall be provided with an aerator.

3.6.1.7.1 Globe valves. Globe valves shall be rated for hot or cold water pressures up to 125 psi (861.75 kPa). Valves shall be of composition bronze and shall have a removable disc-holder with replaceable discs suitable for water temperatures up to 212° Fahrenheit (F) (100° Centigrade (C)). Valve bodies shall have threaded ends. The working pressure shall be cast on the side of the valve body as acceptable evidence of compliance with pressure requirements.

3.6.2 Type II kettles. Type II kettles shall conform to the dimensional requirements of table IV. The two (twin) 5-gallon (18.9 L) capacity kettles shall consist of two single 5 gallon (18.9 L) capacity kettles as specified hereinafter except that each of the 5-gallon (18.9 L) capacity kettles shall be capable of being operated independently and shall be individually controlled by a temperature control (see 3.6.6.2) and safety cutout (see 3.6.6.6).

3.6.2.1 Kettle body. The body shall be cylindrical with either a hemispherical or sloping bottom and shall be fabricated from type 302 or 304 stainless steel as specified in 3.4.1. The material thickness of that section of the body forming the inner wall of the steam-jacket shall conform to the requirements of the ASME Code for the rated working pressure of the kettle. The body shall be seamless drawn or spun from one piece of material. The top edge shall be formed into not less than a 5/16 inch (7.9 mm), half round, open radius, a fully closed sealed bead, or a double hem. A V-shaped pouring lip either formed as a part of the body or welded to the top edge shall be provided. The rim of the body shall fit the flange on the cover.

3.6.2.2 Steam-jacket. The steam-jacket shall be stainless steel. Material thickness shall conform to the requirements of the ASME Code for the rated working pressure of the kettle. The steam-jacket shall be formed or spun from one piece or shall be fabricated by welding preformed segments. The top of the steam-jacket shall be welded to the kettle body. A means for complete drainage of the steam-jacket, either through the safety valve connection or a drain plug shall be provided.

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TABLE IV. Dimensions (type II).

Measurements	2-1/2 gallons (9.5 Liters)	5 1/ gallons (18.9 Liters)	10 gallons (37.9 Liters)
	Inches (mm)	Inches (mm)	Inches (mm)
Overall outside diameter at kettle top (max)	13 (330.2)	17 (431.8)	19 (482.6)
Height from bottom of kettle mount to kettle rim (max)	21 (533.4)	23 (584.2)	28 (711.2)
Clearance between spout (in full tilt position) and top of stand (min)	4-1/2 (114.3)	4-1/2 (114.3)	4-1/2 (114.3)

1/ Applicable to each 5-gallon (18.9 L) kettle.

3.6.2.3 Tilting handle. The tilting handle shall consist of a stainless steel rod and a heat resistant plastic knob. The handle shall be positioned on the right hand side when facing the kettle front and the rod shall be inserted into the trunnion shaft or into a lug welded to the kettle body above the steam-jacket. The knob shall be a push-on type or be threaded to receive the threaded end of the rod.

3.6.2.4 Kettle mountings. Each kettle shall be provided with either one or two trunnions welded to the kettle body. When one trunnion is provided, the kettle shall be mounted on a combination kettle support and tilting trunnion. The combination support shall be fabricated of stainless steel or cast aluminum finished as specified in 3.9. When two trunnions are provided, the trunnion shafts shall be welded to the body above the steam-jacket and the trunnion bushings shall be mounted on two stainless steel leg supports. The kettle provided with a combination support shall be equipped with a stainless steel base plate. Both types of kettle mountings shall have provisions for bolting the kettle assembly to the stand or table specified in 3.6.2.7 or to a counter top. The trunnion(s) shall allow 95° (1.65 rad) of tilt from the normal position and shall allow the kettle to return to the upright position when released. The trunnions and supports shall show no visible signs of deformation when tested as specified in 4.5.3.

3.6.2.5 Cover. The kettle shall be provided with a one-piece removable cover, formed from type 302 or 304 stainless steel as specified in 3.4.1 and shall be a nominal 0.050 inch (1.27 mm) thick or thicker. The cover shall be

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flanged, dished, or suitably reinforced to prevent distortion as a result of normal use and shall be provided with a bar-type loop, or knob type handle. The cover shall fit snugly to the rim of the kettle body.

3.6.2.6 Wire basket. Unless otherwise specified (see 6.2), a basket fabricated of stainless steel wire or a perforated basket fabricated of stainless steel shall be furnished. The basket shall fit the inside contour of the kettle body and shall be provided with two handles. Spacing between wire on the wire baskets shall be not greater than 3/8 inch (9.52 mm). Holes in the perforated baskets shall be not less than 1/8 inch (3.18 mm) in diameter, and unless otherwise specified (see 6.2), shall be not greater than 1/2 inch (12.7 mm) in diameter. A contour-fitted nylon bag with stainless steel spreader-ring shall be furnished with each basket.

3.6.2.7 Tables and stands.

3.6.2.7.1 Table. When specified (see 6.2), the type II kettle shall be provided with a table in accordance with the applicable style and size of RR-T-25.

3.6.2.7.2 Stand. When specified (see 6.2), the type II kettles shall be provided with a stainless steel open stand. Except for a type II, size 8 kettle, the stand shall not require a floor space greater than 28 inches (711.2 mm) long (side to side) by 34-1/2 inches (876.3 mm) deep (front to back) and the combined height of kettle and stand shall be not greater than 53 inches (1346.2 mm). For a type II, size 8 kettle, the length dimension (side to side) will not require a floor space greater than 38 inches. All other dimensional requirements remain the same. The frame shall be stainless steel barstock or round or square tubing with provision for bolting the kettle to the frame. When two kettle trunnion supports are provided, the stand shall be provided with a sheet metal stand top and supporting frame; and when a combination trunnion-and-kettle support is provided, the base shall be utilized as the stand top. The stand shall have provisions for supporting a container to receive kettle contents and shall be provided with a trough or other suitable drain for catching overflow and spillage. The legs shall have adjustable feet with provisions for bolting to the floor. The stand shall show no visible signs of deformation when tested as specified in 4.5.3.

3.6.3 Safety valve. Each kettle shall be provided with an attached pressure-type, brass safety valve constructed in accordance with applicable requirements of the ASME Code. The valve shall be set at the factory in accordance with the ASME Code. The valve shall be provided with a reversible lifting lever and shall be capable of being rotated 360° (6.3 rad) without disturbing the valve setting. The valve shall be provided with a bottom inlet and a side discharge and shall be connected to the steam-jacket. The valve shall be finished as specified in 3.9.

3.6.4 Pressure gauge. A dial type pressure gauge shall be provided and shall conform to the applicable requirements of the ASME Code and shall be located in an opening in the control housing specified in 3.6.6.7 so that the dial is visible. The gauge shall be protected from steam by an internal syphon or coil-pipe syphon below the gauge.

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3.6.5 Water level indicator. Each kettle shall be equipped with a device to indicate when the water in the steam-jacket is at a low level. Low level shall be the point at which there is still sufficient water in the jacket to cover the heating elements. The water level indicator shall be located in a position to be visible to the attendant.

3.6.6 Electrical components.

3.6.6.1 Heating elements. The kettles shall be provided with removable immersion heating elements conforming to the electrical characteristics specified in 3.5.3 and table II. The heating units shall be equipped with elements consisting of helically coiled resistance wire embedded in a magnesium oxide powder within a copper sheath. Sheaths shall be brazed to a brass screw plug or bolted to a flanged type header and the terminals shall be sealed with a moisture-barrier sealing material. The terminals shall be enclosed in the control housing specified in 3.6.6.7.

3.6.6.2 Temperature control. Each kettle shall be equipped with a device to automatically control temperature in the steam-jacket when the selected temperature is attained (see 4.5.2). The temperature control shall have a minimum operating range of 150° to 265°F (65.6° to 129.4°C) and shall be marked with an "OFF" position or have a separate "ON-OFF" switch. An indicating light shall show when the heating elements are energized and may be an integral part of, or be separate from, the temperature control.

3.6.6.3 Contactors. Contactors shall be current rated for resistance heating. When multiple contactors are provided they shall be wired to divide the load. The contactors shall conform to applicable requirements of NEMA ICS 1 and NEMA ICS 2. Unless otherwise specified (see 6.2), contactors shall be a 3- or 4-pole magnetic type, rated up to 600V per pole, with holding coils for 50-60 Hz, and a range of 208 to 230V.

3.6.6.4 Low water cutoff. Each kettle shall be provided with a low water cutoff. The cutoff may be either a float type or electrically-operated probe and relay type and shall conform to NEMA ICS 1 and NEMA ICS 2. The cutoff shall be designed to operate on the water level in the steam-jacket and when tested as specified in 4.5.2 shall automatically cut off the electric current to the elements when the water level in the water level indicator drops to the "LOW" mark. The low-water cutoff may be either a manual reset or automatic reset type. The float chamber of the float type cutoff shall be separated from the switch and operating mechanism by a bellows or other packless device. The unit shall be rated for a maximum 40 psi (275.8 kPa) pressure. The switch electrical contacts shall be enclosed in the control housing specified in 3.6.6.7. The relay of the probe and relay type cutoff shall be mounted on the kettle or be suitable for wall mounting.

3.6.6.5 Pressure limit control. The type I kettle shall be equipped with a device to de-energize the circuit to the heating elements when pressure exceeds the rated working pressure of the kettle (see 4.5.2).

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3.6.6.6 Safety cutout. The type II kettle shall be equipped with a device to de-energize the circuit to the heating elements when the kettle is tilted. Each of the two (twin) 5 gallon kettles shall operate independently.

3.6.6.7 Control housing. The steam-pressure control, pressure gauge, water level indicator, low-water cutoff, heating element terminals, and other necessary controls shall be located in a protective housing. The housing shall be formed as an integral part of the outer casing or may be formed separately. Material for the housing shall be stainless steel as specified in 3.4.1. The housing shall be provided with a hood and shall consist of a top, front, and sides designed to prevent spillage from the kettle from entering the housing. When the housing is not an integral part of the casing, the housing shall be attached to the casing by welding or stainless steel sheet metal screws. A moisture and water repellant gasket shall be installed between the housing and kettle outer casing along the top and two sides when the housing is secured by screws. A removable control housing panel or cover to allow free access to the controls shall be provided.

3.7 Data name plates. Data name plates shall be made of minimum 20 gauge corrosion-resisting metal and attached to each item in such a manner as to meet the applicable sanitary requirements for this equipment. The plate shall contain the following information stamped, engraved or applied by photosensitive means:

National Stock Number

Procurement Instrument Identification Number

Manufacturer's Name, Address, Phone Number

Manufacturer's Model Number

Each plate shall be placed so that it is readily visible to the operator during normal operating use. Each plate shall be placed in a manner so as to not adversely affect the life and utility of the item.

3.8 Instruction plate. Each kettle shall be equipped with a user's instruction plate constructed of not less than 0.025 inch (0.64 mm) thick corrosion-resistant metal permanently fastened to the outside front. The plate, shall contain the following information etched, embossed, or die stamped thereon in legible characters. The words "STEAM JACKETED KETTLE" shall be not less than 3/16 inch (4.76 mm) high and the words "USER's INSTRUCTIONS" shall be not less than 1/8 inch (3.18 mm) high. All other characters shall be not less than 3/32 inch (2.39 mm) high.

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STEAM-JACKETED KETTLE

USER'S INSTRUCTIONS

1. Do not tamper with or obstruct safety valve.

2. Place main disconnect switch in "OFF" position before washing kettle.

3. Wash kettle before and after each use with hot soapy water and rinse inside thoroughly with clean hot water. Do not use hose on outside of kettle. Wash outside of kettle thoroughly without allowing water to enter into control housing. Dry thoroughly inside and outside.

4. Leave cover open when not in use.

5. Filling of steam-jacket shall be done through the connection to the safety valve, or through the air vent (plate shall contain only applicable portion). Hold safety valve open when filling, to release air in jacket.

6. Never energize elements unless level of water recommended by manufacturer is in the steam-jacket.

7. Follow manufacturer's air venting instruction.

8. Clean strainer, draw-off valve and draw-off tube after each use in accordance with manufacturer's recommendations.

3.9 Finish. Kettles shall be finished in accordance with NSF No. 4. When provided, stands and tables shall be finished in accordance with NSF No. 2. The safety valve and other exposed metal components and fittings not fabricated of stainless steel or aluminum shall be nickel-chrome-plated in accordance with the manufacturer's standard commercial practice.

3.10 Wiring diagrams. The contractor shall provide wiring diagrams for each kettle provided, showing the necessary connections for the contactors, temperature control, low-water cutoff, and heating elements.

3.11 Filling instructions. The contractor shall provide filling instructions, normally provided in commercial practice, including exact amount of water required for refilling.

3.12 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

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3.13 Workmanship. All components and assemblies of the kettles shall be free from dirt and other extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings and molded parts if used, shall be free of sand, fins, pits, blowholes, and sprues. External surfaces shall be free from sharp edges and sharp corners.

3.13.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.13.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice and all bolts, nuts, and screws shall be tight.

3.13.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.13.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation of failure when the parts connected by the weld are subjected to proof and service loadings.

3.13.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, not does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.1.3 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standard(s) cited in 3.3. The Government reserves the right to examine and test all kettles to determine the validity of the certification.

4.1.3.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL listing mark, or label; or certified test report from a recognized independent testing laboratory indicating that the kettles have been tested and conform to UL 197.

4.1.3.2 ASME. Acceptable evidence of meeting the applicable requirements of the ASME Code shall be one of the following:

a. A certificate stating that the jacketed sections of the kettles bear the ASME stamp.

b. A certified test report from a recognized independent testing laboratory acceptable to the Government certifying that the jacketed sections of the kettles have been tested and conform to all the applicable requirements of the ASME Code.

4.1.3.3 NSF. Acceptable evidence of meeting the requirements of NSF shall be one of the following:

a. Listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished kettles and stands or tables.

b. A certification for the kettles and stands or tables issued by NSF under their one-time contract evaluation/certification service.

c. A certified test report, acceptable to the contracting officer with the advice of the Army Surgeon General, from an independent testing laboratory indicating that the items have been tested and conform to the specified NSF Standard No. 2 and No. 4.

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4.1.3.4 NEMA. Acceptable evidence of meeting the requirements of NEMA shall be a certificate of compliance stating that the magnetic contactors utilized in the kettles conform to the applicable requirements of NEMA ICS 2.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one kettle when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5 and when specified, the preproduction pack inspection of 4.6 (see 4.6 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one complete machine of the type required. All machines offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for dimensional examination. Dimensional examination shall be based on inspection level S-2 and an Acceptable Quality Level (AQL) of 4.0 percent defective.

4.3.2 Sampling for visual examination. Visual examination shall be based on inspection level II and an AQL of 2.5 percent defective.

4.4 Examination. Each kettle shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual

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examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4.1 Dimensional measurement examination. Each sample shall be examined for the dimensional measurement requirements in section 3 of this specification. Any dimension not within the specified requirement shall constitute one defect.

4.4.2 Visual examination. Each kettle shall be examined for defects listed in table V. Each attribute within each classification of multiple defects shall constitute one defect.

TABLE V. Classification of defects.

Classification	Defect	Requirements Paragraph
MAJOR		
101	Not manufacturer's standard product.	3.1
102	Any wearing part not accessible for adjustment or replacement.	3.5.4
103	Any component missing.	3.6
104	Any component fractured or malformed.	3.6
105	Any component misplaced or not in alignment.	3.6
106	Functioning component that is inoperative or will not operate as intended.	3.6
107	Functioning component that required abnormal force to operate.	3.6
108	Any component not in proper alignment.	3.6
109	Any component loose, other than threaded fasteners.	3.6
110	Any adjustable assembly that cannot be adjusted.	3.6
111	Any adjustable assembly not properly adjusted to perform the intended function.	3.6
112	Canopy does not slope away from kettle body.	3.6.1.1

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TABLE V. Classification of defects. (continue)

Classification	Defect	Requirements Paragraph
MAJOR (Continued)		
113	Body not formed as specified.	3.6.1.1 and 3.6.2.1
114	Top edge not as specified.	3.6.1.1 and 3.6.2.1
115	Top of kettle body does not fit cover flange.	3.6.1.1 and 3.6.2.1
116	Drainage provisions missing or not as specified on steam-jacket.	3.6.1.2 and 3.6.2.2
117	Installation of insulation for Type I not as specified.	3.6.1.3
118	Casing not secured to canopy as specified.	3.6.1.3
119	Legs not type or number specified.	3.6.1.4
120	Cover not flanged, dished, or suitably reinforced.	3.6.1.5 and 3.6.2.5
121	Cover missing.	3.6.1.5.1, 3.6.1.5.2, and 3.6.2.5
122	Cover hinges not as specified.	3.6.1.5.1 and 3.6.1.5.2
123	Cover stops not provided as specified (two-piece cover).	3.6.1.5.2
124	Sanitary shield not located on cover as specified (two-piece cover).	3.6.1.5.2
125	Back section of cover not flanged up (two-piece cover).	3.6.1.5.2
126	Cover handle missing or not as specified.	3.6.1.5.3 and 3.6.2.5

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TABLE V. Classification of defects. (continue)

Classification	Defect	Requirements Paragraph
MAJOR (Continued)		
127	Draw-off tube opening not located as specified.	3.6.1.6.1
128	Draw-off tube not fabricated as specified.	3.6.1.6.1
129	Draw-off tube not flush with opening in valve.	3.6.1.6.1
130	Recesses in draw-off tube which might retain food particles.	3.6.1.6.1
131	Strainer in hook not perforated as specified.	3.6.1.6.1
132	Strainer in hook stay not snug fit in outlet.	3.6.1.6.1
133	Strainer in hook handle not as specified.	3.6.1.6.1
134	Draw-off valve not compression-disc type.	3.6.1.6.2
135	Draw-off valve handle not type specified.	3.6.1.6.2
136	Draw-off valve not removable without tools.	3.6.1.6.2
137	Swing spout bracket valve missing (if valve is not part of spout).	3.6.1.7
138	Swing spout globe valve missing (if valve is not part of spout).	3.6.1.7.1
139	Pouring lip on Type II kettle body not as specified.	3.6.2.1
140	Type II Kettle mounting not type specified.	3.6.2.4
141	Trunnion will not allow 95° tilt.	3.6.2.4
142	Kettle does not return to upright position.	3.6.2.4

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TABLE V. Classification of defects. (continue)

Classification	Defect	Requirements Paragraph
MAJOR (Continued)		
143	Stand frame not of specified material (tubing or bar stock).	3.6.2.7.2
144	Stand container support not adequate.	3.6.2.7.2
145	Stand drain missing or not as specified.	3.6.2.7.2
146	Safety valve not capable of 360° rotation.	3.6.3
147	Safety valve not connected to steam-jacket as specified.	3.6.3
148	Safety valve not provided with bottom inlet and side discharge.	3.6.3
149	Pressure guage not as specified.	3.6.4
150	Water level indicator not as specified.	3.6.5
151	Heating elements not removable.	3.6.6.1
152	Heating element missing or damaged.	3.6.6.1
153	Heating element terminals not sealed.	3.6.6.1
154	Temperature control not as specified.	3.6.6.2
155	Temperature control indicator light missing.	3.6.6.2
156	Pressure limit control missing or not as specified.	3.6.6.5
157	Safety cutout not provided with type I kettle.	3.6.6.6
158	Control housing access not as specified.	3.6.6.7
159	Instruction plate missing, incomplete, or not legible.	3.8

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TABLE V. Classification of defects. (continue)

Classification	Defect	Requirements Paragraph
MAJOR (Continued)		
160	Type of finish not as specified.	3.9
161	Finish is peeling, chipped, or scratched (plating).	3.9
162	Not finished where required.	3.9
163	Any burr, sliver, sharp edge, rough die, tool, gauge, trim or grind marks.	3.13
164	Any component fractured, buckled, bent, dented, punctured, or deformed.	3.13
165	Threaded fasteners missing, broken, stripped, or fractured.	3.13.2
166	Threaded fasteners loose.	3.13.2
167	Rivets missing, broken, or inadequately riveted.	3.13.3
168	Rivets loose or not in full contact with members riveted.	3.13.3
169	Welding or soldering not free from rust, scale, paint, grease, and other foreign matter.	3.13.4
170	Welds not continuous, smooth and free of cracks, burn holes, undercuts, fissures or not completely fused.	3.13.4
171	Finished weld area shows evidence of scale.	3.13.4
172	Welding and soldering flux not removed.	3.13.4
173	Welds or soldering missing, not adherent, or incomplete.	3.13.4
174	Welding or soldering not clean (flux or flux residue not removed), not smooth, or having pinholes.	3.13.4

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TABLE V. Classification of defects. (continue)

Classification	Defect	Requirements Paragraph
MAJOR (Continued)		
175	Castings not sound, has patchings, misplaced coring, warping or other defects.	3.13.5
MINOR		
201	Handle loose.	3.6.1.5.3
202	Swing spout aerator missing.	3.6.1.7
203	Data name plate marking missing, incomplete, or illegible.	3.7
204	Instruction plate not specified type or not in proper location, or illegible.	3.8

4.5 Tests.

4.5.1 Capacity. The kettle shall be tested to determine compliance with capacity requirements of 3.5.1 or 3.5.2, as applicable. The test shall be performed by filling the kettle with water from a container of known capacity. Any nonconformance with specified requirements shall constitute failure of this test.

4.5.2 Operation. The kettle shall be tested to determine conformance with the operational requirements of 3.6.6.2, 3.6.6.4, 3.6.6.5, and 3.6.6.6 as applicable. Any nonconformance with specified requirements shall constitute failure of this test.

4.5.3 Trunnions, support(s) and stand or table. The type II kettle shall be loaded with weights equivalent to 1-1/2 times the weight of its water capacity and the trunnions, support(s), and stand or table when provided shall be checked to determine conformance with the deformation requirements of 3.6.2.4 and 3.6.2.7.2, respectively. Any nonconformance with specified requirements shall constitute failure of this test.

4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5. When specified a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. For type I kettles, the kettle shall be disassembled from the stand, detachable tilting handle removed from the kettle, and the stand completely disassembled when possible. The outlet strainer shall be removed from the type II kettle.

5.1.1.2 Kettle.

5.1.1.2.1 Type I. The cover of each kettle shall be secured in place with tape conforming to PPP-T-60, or type II of PPP-T-97, or cord or twine, in such a manner as to prevent its loosening during transit. The swing-spout and other parts of the kettle, free to move, shall be secured in place to prevent movement during transit. The feet shall be adjusted to the minimum dimension.

5.1.1.2.2 Type II. The wire basket shall be cushioned inside the kettle in such a manner as to prevent movement and abrasion of the kettle. The cover shall be secured in place as specified in 5.1.1.2.1. The kettle body shall be secured in such a manner as to prevent its tilting during transit.

5.1.1.3 Components. The outlet strainer, strainer hook, detached tilting handle, and stand or table, when disassembled, shall be packaged in a fiberboard box conforming to style RSC, type CF or SF, class weather-resistant or domestic of PPP-B-636 (see 5.2.1.1 or 5.2.2.1). Components shall be immobilized and prevented from contacting each other by the use of fiberboard inserts or other cushioning material in such a manner as to prevent damage during transit. Each box shall be securely closed in accordance with the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.6.

5.1.1.4 Stand or table. When provided, the stand or table for type II kettles, when not disassembled, shall have the feet adjusted to the minimum dimension, and all components that are free to move, shall be secured in a fixed position in such a manner as to prevent movement during transit.

5.1.2 Level C. Kettles shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The contractor may use his standard practice when it meets this requirement.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Kettles, 60 gallons (227.1 L) or less. Each kettle, 60 gallons (227.1 L) or less, packaged as specified in 5.1, shall be packed in a shipping container conforming to overseas type of PPP-B-601; or class 2 style 2 of PPP-B-621. A minimum clearance of one inch shall be maintained between the kettle and the interior surfaces of the top and four sides of the shipping container. The fiberboard box containing disassembled components shall be class weather-resistant (see 5.1.1.3), waterproofed by means of tape in accordance with the appendix of PPP-B-636, and secured in an unused portion of the shipping container. Contents shall be secured by wood blocking and bracing or anchoring in accordance with MIL-STD-1186. Each shipping container shall be provided with skids and closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.1.2 Kettles, 80 gallons (302.8 L). Each 80-gallon (302.8 L) kettle, packaged in accordance with 5.1, shall be packed in a closed crate conforming to type I, class 1 or 2, style a of MIL-C-104. The fiberboard box containing disassembled components shall be secured in an unused portion of the crate. The contents shall be anchored and the crate closed and strapped in accordance with MIL-C-104 or the appendix thereto.

5.2.2 Level B packing.

5.2.2.1 Kettles, 60 gallons (227.1 L) or less. Each kettle, 60 gallons (227.1 L) or less, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that shipping containers shall conform to domestic type, style A or B of PPP-B-601, or class 1, style 2 of PPP-B-621; and fiberboard box may be class domestic with waterproofing not required.

5.2.2.2 Kettles, 80 gallons (302.8 L). Each 80-gallon (302.8 L) kettle, packaged as specified in 5.1, shall be packed as specified in 5.2.1.2, except that waterproof requirements for the crate shall be waived.

5.2.3 Level C. Kettles, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Kettles covered by this specification are intended for cooking food in permanent kitchens, hospitals, and galleys.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and size of kettle required (see 1.2, 3.5.1, and 3.5.2).
c. When insulation and outer casing are required on the type I kettles for shipboard use (see 3.5.1).
d. When type II kettles are not to be charged at the factory with a permanent supply of water containing rust inhibitors and any other necessary additives (see 3.5.2).
e. When electrical characteristics are other than as specified (see 3.5.3 and table II).
f. When electrical characteristics are to be designed for Navy shipboard use (see 3.5.3 and table II).
g. When the body and steam-jacket are to be enclosed by a cylindrically shaped casing (see 3.6.1.3).
h. When the diameter and size of the draw-off tube and draw-off valve for type I kettles are to be nominal 3-inch (see 3.6.1.6).
i. When the swing-spout water supply connection for the type I kettle is not required (see 3.6.1.7).
j. When a basket is not required for type II kettles. When holes in perforated basket are to be greater than 1/2 inch (see 3.6.2.6).
k. When a table is required for type II kettles (see 3.6.2.7.1).
l. When a stand is required for type II kettles (see 3.6.2.7.2).
m. When contactors are to be other than as specified (see 3.6.6.3).
n. When a first article sample and inspection is required. When a preproduction pack inspection is required and time frame required (see 4.2.1 and 4.6).
o. Level of preservation and packaging (see 5.1).
p. Level of packing (see 5.2).
q. When additional special marking is required on interior packages and shipping containers (see 5.3).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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6.5 Cross-reference of classification. The classification of the kettles in this revision has been changed from the previous revision of this specification. The following list reflects the changes in the classification of this revision and the classification of the previous revision.

MIL-K-43359E	MIL-K-43359D
STYLES	STYLES
Deleted	Style A
Deleted	Style B
Deleted	Style C

6.6 Subject term (key word) listing.

Appliance, cooking
Appliance, commercial
Food service equipment
Kettles, steam-jacketed, electrically heated

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - YD

Review Activities:
Navy - MC, MS, SA
DLA - GS

User Activity:
Navy - SH

Preparing Activity:
Navy - YD

(Project 7310-0708)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-K-43359E	KETTLES, STEAM-JACKETED (STAINLESS STEEL), ELECTRICALLY HEATED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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